

Kadhiravan Gopal

Chicago, IL | (312) 730-8530 | iitzkadhiravan@gmail.com | linkedin.com/in/kadhiravang | github.com/kadhiravang | kadhiravang.github.io

SUMMARY

AI/ML and full-stack software engineer with 2 years in LTIMindtree's Generative AI Center of Excellence, working on 8+ full-stack applications (Python backend, Angular UI) and responsible for them from development through deployment and production support. Hands-on with LLMs (GPT-4o, Claude, Mistral, Amazon Bedrock, Azure OpenAI), RAG, multi-agent systems, vector stores, and evaluation. MS in Artificial Intelligence at Illinois Institute of Technology (GPA 3.77/4.0, graduating Dec 2026); seeking full-time AI/ML engineering roles.

WORK EXPERIENCE

Software Engineer, Generative AI Center of Excellence — LTIMindtree Jul 2022 – Jul 2024

- Worked on 8+ full-stack applications (Python backends, Angular UIs); trusted to start projects from scratch, and responsible for development, debugging, deployment, and live production support.
- Primary developer of an AI-assisted property insurance underwriting assistant, built from scratch with a small agile team. It accepted audio, video, PDF, Word, presentation, and image inputs; I wrote the multi-agent architecture (manual orchestration, before LangGraph and native function calling existed) that identified each file, processed files in parallel, and produced a property estimate, cutting per-property underwriting from 1–1.5 months of manual work to minutes.
- Raised RAG accuracy from 70% to 95% on our test cases by researching and implementing LLM-based chunk re-ranking.
- Stack: Angular, Flask REST APIs, MongoDB, Qdrant, LlamaIndex, LangChain; evaluated GPT-4o, Claude, Mistral, and others. Deployed GPT-4, GPT-4o, and GPT-3.5 Turbo on Azure OpenAI (Key Vault, AI Search, Functions, Document Intelligence) and apps on Azure App Service and AWS.
- On a client project limited to one Azure OpenAI deployment, negotiated a realistic target (8+ simultaneous users, 1–2 minute responses) and built a “canvas” layer that screened malicious or violent requests, tracked requests and tokens per user for usage and pricing, and queued requests to stay within model limits.
- Fixed live production issues (including night incidents); ran knowledge-transfer sessions; presented a handwriting assistant and other apps at IgnAlte and internal demos.

Student Assistant, CPS Testing Project — Illinois Institute of Technology Oct 2025 – Present

- Rotate across admin, check-in, and proctoring roles; as admin, collect student and session data, enter it into the testing software, and track it across multi-campus teams.

PROJECTS

HaloCheck — LLM Evaluation & Hallucination Detection (private academic project) 2026

Evaluation harness for LLM-generated clinical summaries using a DeBERTa cross-encoder NLI, scispaCy, medspaCy, and NegEx for entity extraction, negation detection, and severity-weighted factual-fidelity scoring, with regression tracking. Outperformed a published baseline (HHEM): F1 0.64, Recall 0.83, OOD F1 0.93.

Agentic RAG Bot — LangGraph, FastAPI, Qdrant, React 2026

Session-isolated RAG over PDFs: a LangGraph agent rewrites, retrieves, grades, and retries; every claim is cited to document, timestamp, and page, and uncovered questions return “not covered” instead of a guess.

FitnessKitchen — AI Nutrition Tracker (Android, Kotlin, Jetpack Compose) 2026

Built end to end: logs meals from natural-language text or voice, using Gemini function-calling to ground each entry in USDA nutrition data.

Mini-CLIP and Mini-DiT — Paper Reimplementations in PyTorch (coursework) 2025–2026

Reimplemented CLIP (custom training loop, contrastive loss, zero-shot evaluation, bias/fairness analysis; ~72% top-1 on mini-scale ImageNet) and the Diffusion Transformer paper (ICCV 2023) across MNIST, BDD100K, and CIFAR-10.

TECHNICAL SKILLS

LLMs & Agentic AI: GPT-4o, Claude, Mistral, Amazon Bedrock, Azure OpenAI, Gemini, LangChain, LlamaIndex, LangGraph, RAG, LLM re-ranking, multi-agent orchestration, function calling, prompt engineering, LLM evaluation

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, diffusion models, computer vision, DeBERTa, scispaCy, medspaCy

Engineering: Python (Flask, FastAPI), REST APIs, SQL, Angular, React, Kotlin, Jetpack Compose · PostgreSQL, Oracle, MongoDB, Qdrant, Neo4j (prototype)

Cloud & DevOps: Azure (OpenAI, App Service, Key Vault, AI Search, Functions, Document Intelligence), AWS (Bedrock, ECS, EC2, S3), Docker, CI/CD, Git

EDUCATION

M.S., Artificial Intelligence — Illinois Institute of Technology, Chicago, IL Jan 2025 – Dec 2026

GPA 3.77/4.0 · Machine Learning, Deep Learning, NLP, Medical Informatics & AI, Computer Vision, Algorithms, Databases

B.E., Computer Science — Panimalar Engineering College (Anna University) · GPA 8.72/10 2018 – Jun 2022

CERTIFICATIONS

Oracle SQL Certified Associate · AWS Partner Accreditations (Technical, Cloud Economics, Sales) · CITI Biomedical Research